

Experiment 4:

Practical Exercise: Build and Run a Java Application with Maven, Migrate the Same Application to Gradle.

Step 1: Install Maven

1. Update package index:

```
sudo apt update
```

2. Install Maven:

```
sudo apt install maven
```

3. Verify Maven installation:

```
mvn -v
```

Step 2: Create a Maven Project

1. Open a terminal and run:

```
mvn archetype:generate /  
-DgroupId=com.example /  
-DartifactId=MyMavenProject /  
-DarchetypeArtifactId=maven-archetype-quickstart /  
-DinteractiveMode=false
```

2. Navigate into the project directory:

```
cd MyMavenProject
```

Step 3: Open pom file

```
nano pom.xml
```

Add the following/w </dependencies> and </project>

```
<build>  
<plugins>  
<plugin>  
<groupId>org.apache.maven.plugins</groupId>  
<artifactId>maven-compiler-plugin</artifactId>  
<version>3.8.1</version>  
<configuration>  
<source>17</source>  
<target>17</target>  
</configuration>  
</plugin>  
<plugin>  
<groupId>org.apache.maven.plugins</groupId>  
<artifactId>maven-jar-plugin</artifactId>  
<version>3.2.0</version>  
<configuration>
```

```
<archive>
<manifest>
<mainClass>com.example.App</mainClass>
</manifest>
</archive>
</configuration>
</plugin>
</plugins>
</build>
</project>
```

Step 4:Run on terminal:

`mvn clean install`

Step 5:Run on terminal:

`java -jar target/ MyMavenProject-1.0-SNAPSHOT.jar`

Step 6: Migrate to a Gradle Project

Initialize a Gradle project

`gradle init`

Select implement language:

1:Kotlin

2:Groovy

Enter selection (default: Kotlin) [1..2] 2

Open build.gradle and add

```
plugins {
    id 'application' // Apply the application plugin
}
```

Add the following at the end in build.gradle

```
application {
    mainClassName 'com.example.App'
}
```

Step 7:Run Gradle Commands

gradle build

gradle test

gradle run